Weekly Research Journal

Cartographer-Based Real-Time Vehicle Motion Estimation

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F1TENTH Autonomous Vehicle Project

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Abstract

This week established a real-time whole-vehicle motion estimator based on the Cartographer map to base_link pose. Position differences provide map-frame velocity components, speed magnitude, and movement direction; Cartographer orientation provides heading; and projection onto the body axis provides signed forward speed. Manual and remote-controlled tests verified the complete ROS 2 output set. In a representative 15-sample field sequence, moving speed ranged from 0.426 to 0.726 m/s and 13 samples had a direction-heading difference within +/-10 degrees. This establishes responsive, internally consistent output, but not absolute accuracy because no independent ground truth was recorded. The VESC comparison remains preliminary: sign alignment, calibration, synchronized logging, and fusion are not complete.

Weekly contribution

The work progressed from estimator implementation to controlled manual checks and finally to powered indoor driving. Table 1 states what the available evidence supports and avoids treating qualitative agreement as a calibrated accuracy result.

Table 1: Evidence and scope of the week's principal results.

Contribution	Evidence	Scope
Cartographer motion state	Six real-time topics for velocity and orientation	Completed
Powered field operation	0.426-0.726 m/s; 13/15 angular differences within +/-10 degrees	Functional consistency
VESC comparison	Opposite forward sign; comparable order of magnitude	Preliminary only

1 System configuration and test platform

The F1TENTH platform combines a Hokuyo 2D LiDAR, VESC motor controller and IMU, onboard computing, and the ROS 2 Humble software stack. The measured LiDAR and IMU rates were approximately 10 Hz and 50 Hz, respectively. Cartographer maintained the map to base_link transform used by the vehicle-state estimator.

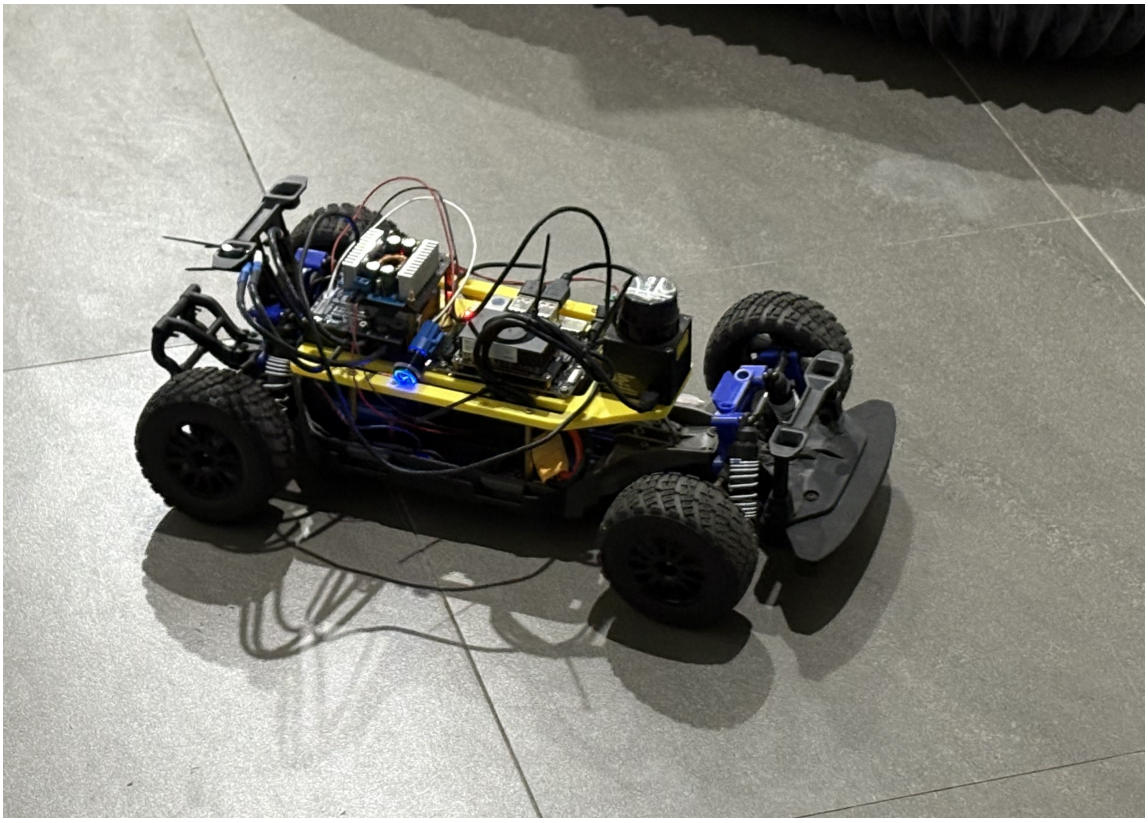


Figure 1: F1TENTH platform during the indoor remote-controlled test. Original color is retained for the physical test record; the crop keeps the vehicle, sensing hardware, and track environment while removing nonessential visual detail.

Table 2: Cartographer-based real-time motion interface.

Topic	Physical quantity	Status
/cartographer_velocity/speed	Speed magnitude	Working
/cartographer_velocity/vx	Map-frame x-velocity	Working
/cartographer_velocity/vy	Map-frame y-velocity	Working
/cartographer_velocity/movement_direction	Velocity-vector direction	Working
/cartographer_velocity/heading	Vehicle yaw	Working
/cartographer_velocity/forward_speed	Signed longitudinal speed	Working

2 Estimation method

2.1 Velocity, movement direction, and heading

For two Cartographer poses (x_1, y_1, t_1) and (x_2, y_2, t_2), map-frame velocity is estimated by finite differences:

$$v_x = (x_2 - x_1)/(t_2 - t_1), \quad v_y = (y_2 - y_1)/(t_2 - t_1). \quad (1)$$

$$v = \mathrm{sqrt}(v_x^2 + v_y^2), \quad \mathrm{phi} = \mathrm{atan2}(v_y, v_x). \quad (2)$$

Cartographer yaw supplies heading theta. Movement direction phi and body heading theta are therefore distinct quantities; their wrapped difference is a useful forward-motion diagnostic, but need not vanish during reverse motion, slip, or lateral repositioning.

2.2 Filtering and signed forward speed

A one-second averaging window and 0.005 m/s deadband suppress localization jitter while retaining slow motion. The earlier 0.025 m/s threshold removed a genuine 0.0184 m/s case. Signed longitudinal speed is obtained by projection onto the body axis:

$$v_{\mathrm{forward}} = v_x \cos(\mathrm{theta}) + v_y \sin(\mathrm{theta}). \quad (3)$$

3 Experimental validation

3.1 Manual and remote-controlled tests

Manual translation produced speed changes from rest through 0.0449, 0.0734, 0.1264, and 0.1859 m/s, and signed forward speed reached approximately +0.29 and -0.33 m/s. The subsequent remote-controlled test exercised simultaneous translation and turning. Figure 2 summarizes 15 consecutive displayed field samples; the final 0.047 m/s point represents deceleration or near-stop behavior and is excluded from the moving-speed range.

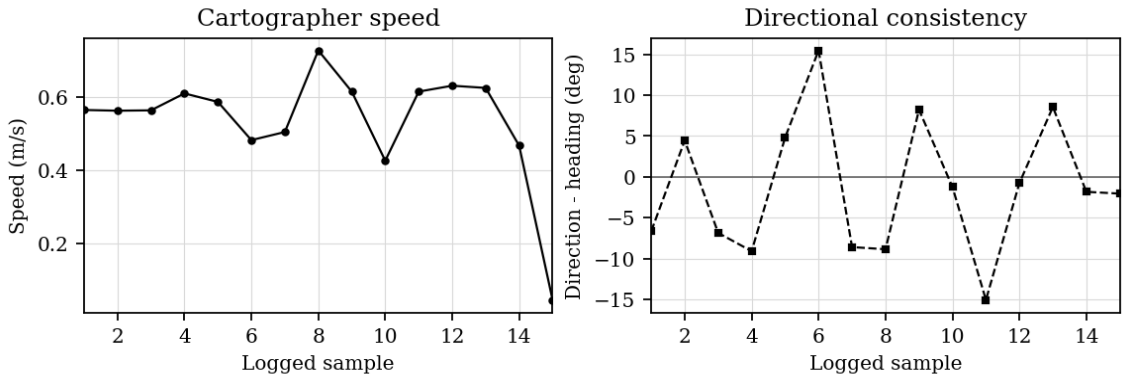


Figure 2: Representative field-test metrics transcribed from the recorded terminal output. The direction-heading difference is within +/-10 degrees for 13 of 15 displayed samples. This is an internal consistency check, not an external accuracy measurement.

3.2 Field-test record and interpretation

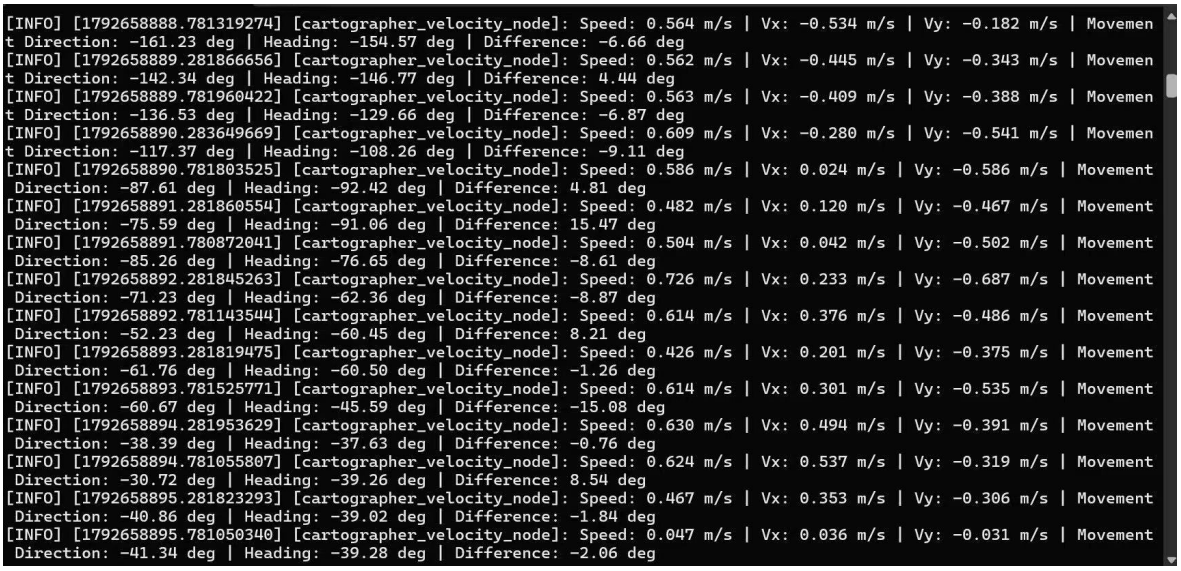


Figure 3: Grayscale record of the real-time Cartographer velocity-node output during remote-controlled driving. The terminal image is retained as evidence; Table 3 provides the legible quantitative summary.

Table 3: Selected samples from the remote-controlled field test.

Speed	v_x	v_y	Movement dir.	Heading	Difference
0.564	-0.534	-0.182	-161.23	-154.57	-6.66
0.609	-0.280	-0.541	-117.37	-108.26	-9.11
0.726	0.233	-0.687	-71.23	-62.36	-8.87
0.630	0.494	-0.391	-38.39	-37.63	-0.76
0.467	0.353	-0.306	-40.86	-39.02	-1.84

4 Preliminary comparison with VESC odometry

VESC odometry uses motor ERPM with the provisional gain 4614.0. In a forward-only test, VESC reported approximately -0.158 to -0.295 m/s while Cartographer forward speed was positive. The present configuration therefore has opposite forward signs. Comparable magnitude is only qualitative evidence because the streams were not recorded as a timestamp-aligned dataset.

Table 4: Current interpretation of the VESC-Cartographer comparison.

Aspect	Observation	Conclusion
Basis	Whole-vehicle displacement versus motor ERPM	Complementary measurements
Sign	Cartographer positive; VESC negative	Alignment required
Magnitude	Both are a few tenths of a metre per second	Not a gain calibration
Timing	No synchronized recorded dataset	No residual statistics yet

5 Completed work and open work

Table 5 separates demonstrated results from planned work. The Cartographer estimator is operational; VESC calibration, synchronization, and fusion are not complete.

Table 5: Project status at the end of the reporting week.

Status	Item
Completed	Real-time speed, v_x, v_y, movement direction, heading, and signed forward speed.
Completed	One-second averaging window and 0.005 m/s low-speed deadband.
Completed	Manual forward/reverse validation and powered remote-controlled test.
Completed	Identification of the VESC/Cartographer forward-sign mismatch.
Pending	Timestamp-aligned logging of ERPM, VESC speed, Cartographer speed, and ground truth.
Pending	Calibration of speed_to_erpm_gain and quantitative residual metrics.
Pending	Sensor-fusion implementation and validation.
Pending	Absolute-accuracy testing against an independent reference.

6 Discussion and next experiment

The principal result is a practical whole-vehicle motion-state interface derived from Cartographer rather than motor rotation. The powered test shows responsive output and broad agreement between motion direction and heading. The evidence establishes functionality and real-time consistency, but not absolute measurement accuracy.

The next experiment should record all signals with ROS timestamps while the vehicle traverses a measured distance in controlled forward and reverse runs. After sign alignment, a regression between raw ERPM and independently measured longitudinal speed can replace the provisional gain. Delay, scale, bias, and uncertainty should be quantified before any VESC-Cartographer fusion is implemented.

Planned validation sequence

Step	Measurement	Decision enabled
1	Synchronized ERPM, VESC speed, Cartographer speed	Estimate delay and sign
2	Measured distance and elapsed time	Calibrate scale and bias
3	Repeated forward/reverse runs	Quantify uncertainty
4	Residuals under turns and slip	Assess fusion feasibility

Appendix A Supplementary transform evidence

Figure 4 records the continuously updated map to base_link translation and orientation used by the estimator. The displayed samples are nearly stationary and therefore show only small variation; they demonstrate transform availability rather than motion accuracy.

```
- Matrix:
 -0.646 -0.762  0.046  4.409
  0.762 -0.647 -0.015 12.634
  0.041  0.026  0.999 -0.116
  0.000  0.000  0.000  1.000
At time 1792659413.109426600
- Translation: [4.408, 12.637, -0.116]
- Rotation: in Quaternion (xyzw) [0.024, 0.003, 0.907, 0.420]
- Rotation: in RPY (radian) [0.026, -0.041, 2.275]
- Rotation: in RPY (degree) [1.477, -2.356, 130.325]
- Matrix:
 -0.647 -0.761  0.046  4.408
  0.762 -0.648 -0.015 12.637
  0.041  0.026  0.999 -0.116
  0.000  0.000  0.000  1.000
At time 1792659414.109562300
- Translation: [4.405, 12.633, -0.116]
- Rotation: in Quaternion (xyzw) [0.024, 0.003, 0.908, 0.419]
- Rotation: in RPY (radian) [0.025, -0.041, 2.276]
- Rotation: in RPY (degree) [1.459, -2.354, 130.418]
- Matrix:
 -0.648 -0.760  0.046  4.405
  0.761 -0.649 -0.015 12.633
  0.041  0.025  0.999 -0.116
  0.000  0.000  0.000  1.000
At time 1792659415.109560400
- Translation: [4.409, 12.627, -0.116]
- Rotation: in Quaternion (xyzw) [0.024, 0.003, 0.907, 0.420]
- Rotation: in RPY (radian) [0.025, -0.041, 2.274]
- Rotation: in RPY (degree) [1.453, -2.373, 130.292]
```

Figure 4: Grayscale Cartographer transform output showing translation, quaternion orientation, roll-pitch-yaw values, and the homogeneous transform matrix.

Table 6: Reproducibility record for the reported configuration.

Parameter	Value	Interpretation
LiDAR rate	approximately 10 Hz	Cartographer range input
IMU rate	approximately 50 Hz	Orientation/motion input
Averaging window	1.0 s	Reduces differentiated-pose noise
Speed deadband	0.005 m/s	Retains verified slow motion
VESC gain	4614.0	Provisional; not calibrated
Ground truth	Not recorded	Absolute accuracy not established